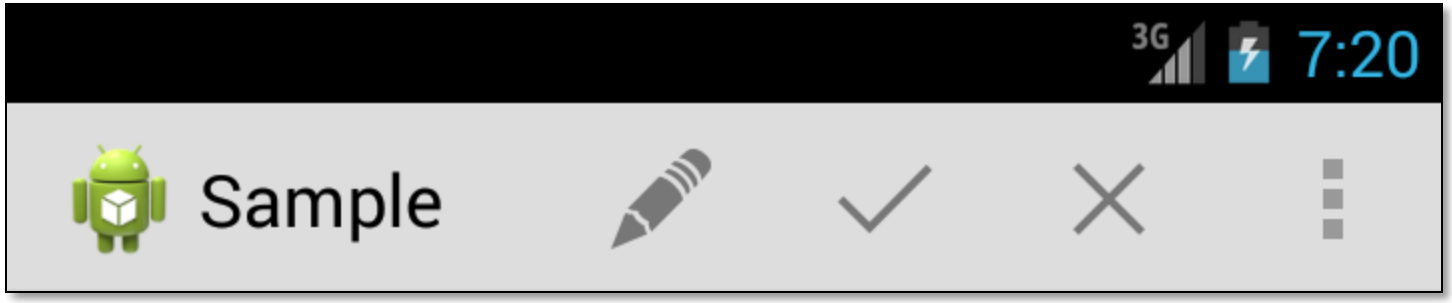
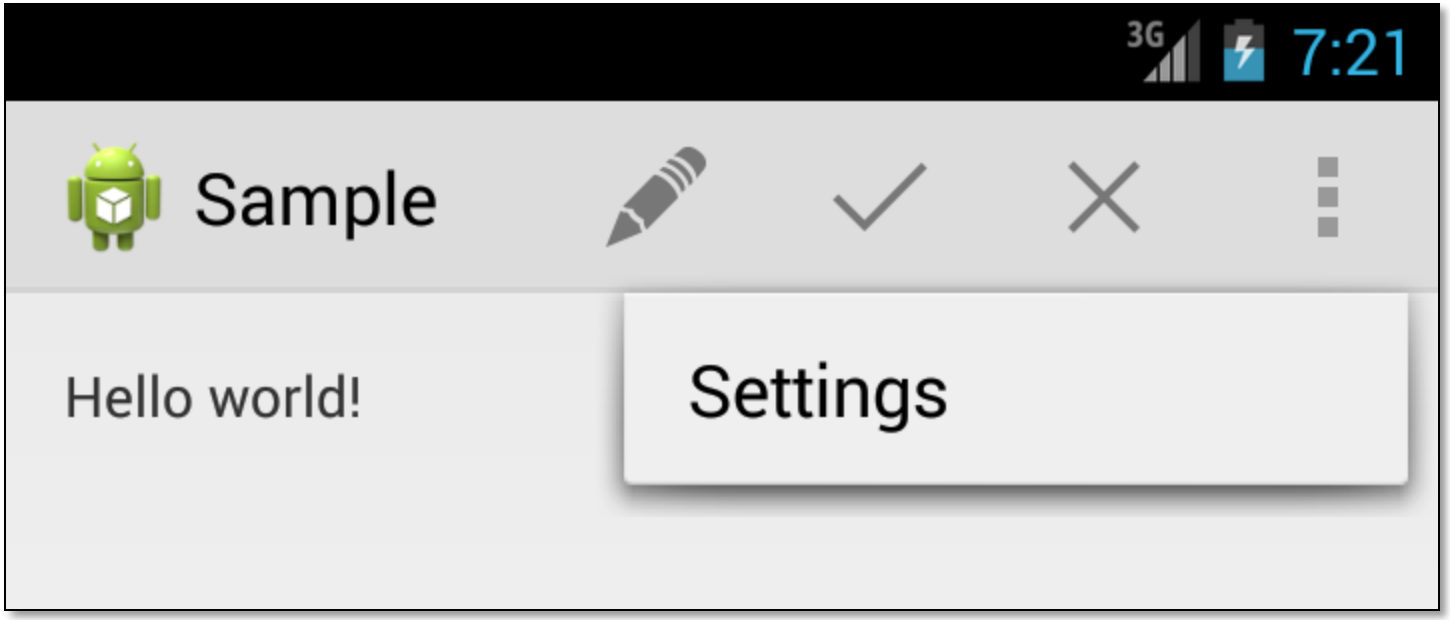
605.486 – Android Development

Module 5a
Action Bars

ActionBar





ActionBar Setup

```androidstudio/res/menu/menu_main.xml
<menu xmlns:android="http://schemas.android.com/apk/res/android"
     xmlns:yourapp="http://schemas.android.com/apk/res-auto" >

    <item
        android:id="@+id/action_new"
        android:orderInCategory="100"
        yourapp:showAsAction="always|withText"
        android:icon="@drawable/ic_action_new_dark"
        android:title="@string/newTodo"/>
    <item
        android:id="@+id/action_settings"
        android:orderInCategory="100"
        yourapp:showAsAction="never"
        android:title="@string/action_settings"/>

</menu>
```

```androidstudio/MainActivity.java
public boolean onCreateOptionsMenu(Menu menu) {
    // Inflate the menu;
    //   this adds items to the action bar if it is present.
    getMenuInflater().inflate(R.menu.main, menu);
    return true;
}

public boolean onOptionsItemSelected(MenuItem item) {
    switch(item.getItemId()) {
        case R.id.action_new:
            // start "new" activity
            return true;
        case R.id.action_settings:
            // display settings
            return true;
        default:
            return super.onOptionsItemSelected(item);
    }
}
```

ActionBar Compatibility

- Google Compatibility Library
- Includes ActionBar Support
- NEEDS v7 appcompat lib
  - Import existing android code to create library project for <SDK>/extras/android/support/v7/appcompat
  - Manifest: use minSdk=7 or later
- See http://developer.android.com/guide/topics/ui/actionbar.html
  - Highlights follow…

ActionBar Compatibility

- Extend ActionBarActivity to use ActionBars
- Use theme (or create a child theme of)

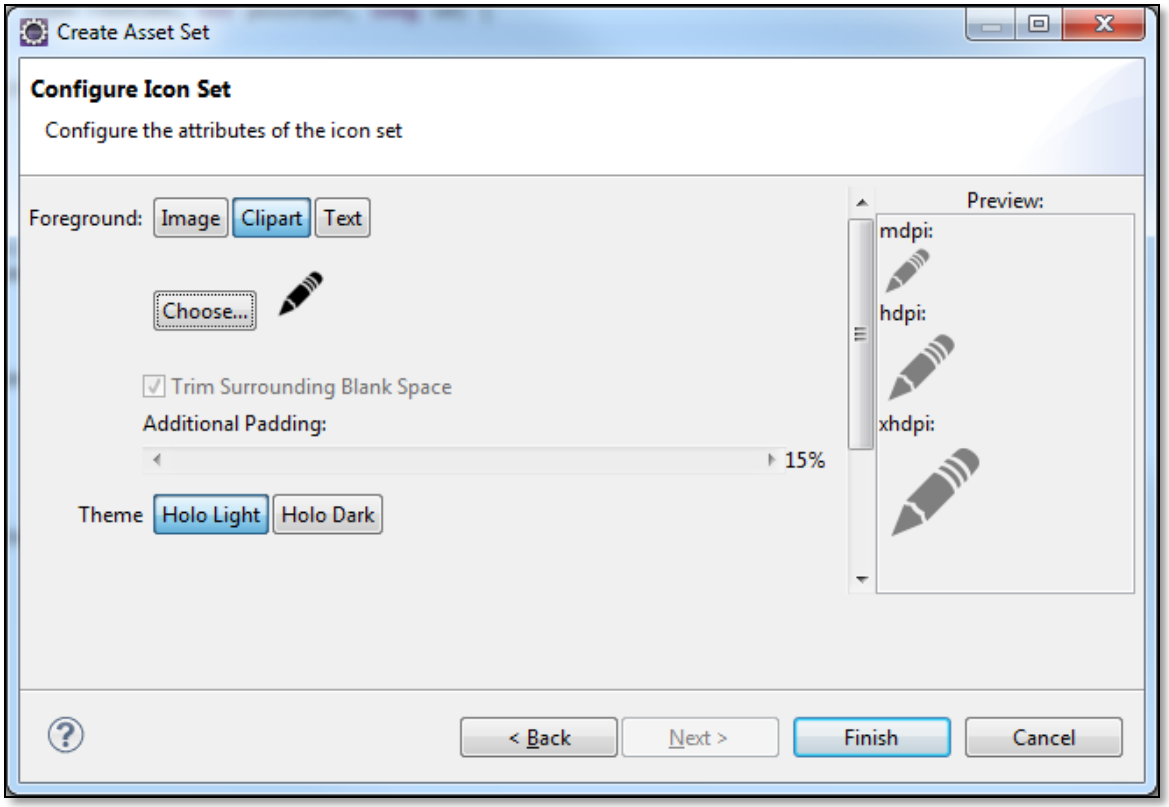
```
<activity
android:theme="@style/Theme.AppCompat.Light" ... >
```

- Menu XML tweaks:

```
<menu  xmlns:android="http://schemas.android.com/apk/res/android"
       xmlns:yourapp="http://schemas.android.com/apk/res-auto"  >

    <item android:id="@+id/action_search“
        …
        yourapp:showAsAction="ifRoom"   />
  …
</menu>
```

Create ActionBar Icons

- New->Android Icon Set
- Example:
  - Edit
  - Done
  - Cancel



ActionBar Home/Up Navigation (1)

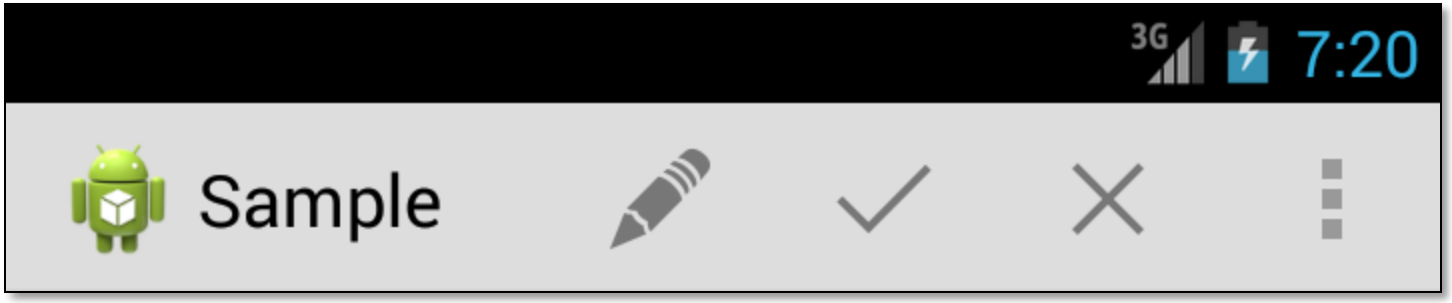
```
public class EditActivity extends ActionBarActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_xyz);
        getSupportActionBar().setDisplayHomeAsUpEnabled(true);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }

}
```

ActionBar Home/Up Navigation (2)

```
<activity
    android:name="com.javadude.ab.EditActivity"
    android:parentActivityName="com.javadude.ab.MainActivity"
    android:label="@string/edit" >
    <meta-data
        android:name="android.support.PARENT_ACTIVITY"
        android:value="com.javadude.ab.MainActivity" />
</activity>
```

Theming for Dark/Light Icons (1)

@style/Theme.AppCompat.Light



@style/Theme.AppCompat.Light.DarkActionBar

Theming for Dark/Light Icons (2)

- Define attribute: res/values/attrs.xml

```
<resources>
    <attr name="actionIconEdit" format="reference"/>
</resources>
```

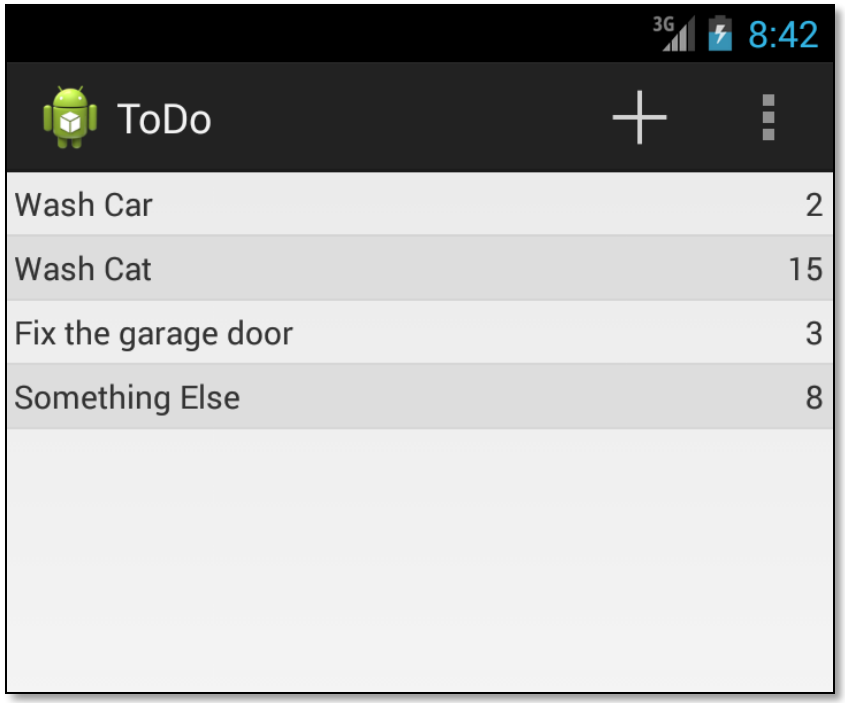
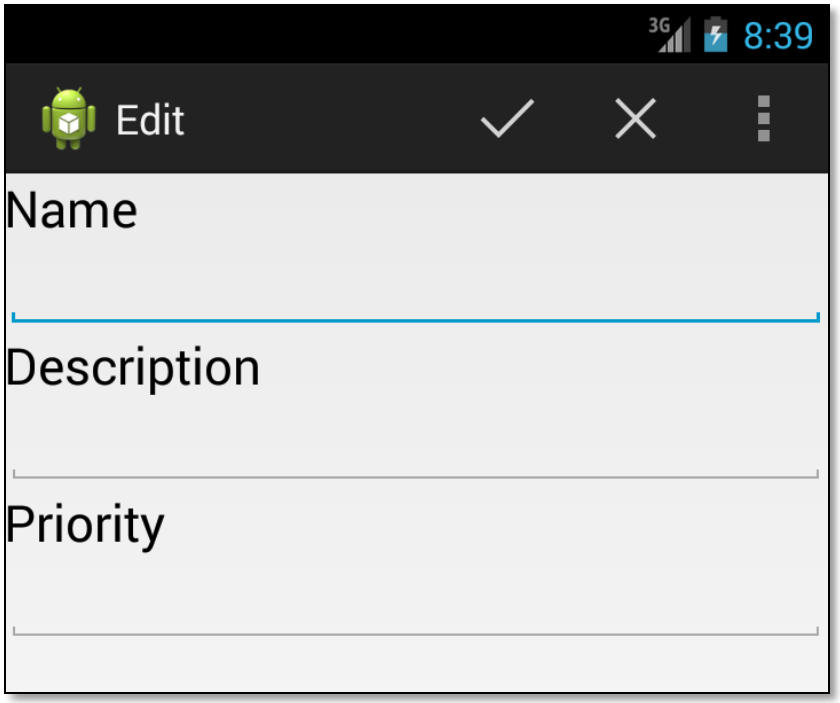
- Create light/dark icons

Theming for Dark/Light Icons (3)

- Define values in styles (change -v11, -v14…)

```
<style name="AppBaseTheme"
        parent="@style/Theme.AppCompat.Light">
    <item name="actionIconEdit">
        @drawable/ic_action_edit_light
    </item>
</style>
```

- Reference in menu

```
android:icon="?attr/actionIconEdit"
```

ToDo List Example